- *Zod schema validation*—Zod is a library for schema declaration and validation. By configuring a Zod schema that defines the expected structure and types of the AI response, we ensure type safety and data integrity, catching any inconsistencies in the AI's output. Zod is used in the Vercel AI SDK as the primary schema validator service.
- *Iterative refinement (reprompting)*—If the previous options fail, we can try to use multiple rounds of AI queries to refine and structure the data. This might improve things, but we must be cognizant of the fact that we will be consuming more resources and tokens to achieve a reliable result.
- *Template-based generation*—Here we provide the AI with a template or example of the desired output structure. Then we ask it to fill in the details. This could improve the consistency in the AI's structured outputs.
- *Postprocessing*—As a final step, we can implement a series of data cleaning and structuring steps after receiving the AI response to ensure data quality and consistency.

In practice, a combination of these techniques is often used to achieve reliable results. The choice of techniques depends on the specific requirements of the application, the capabilities of the AI model being used, and the complexity of the desired data structure. As far as we are concerned, we must aim for reliability, consistency, and compliance if we ever want this to work for our end users. Thus, we need to ensure we use all the available tooling.

### 4.3.3 Tools for implementing type-safe AI-generated content

When working with AI-generated content in a type-safe environment, two key helpers provided by the Vercel AI SDK are particularly useful: `generateObject` and `streamObject`. These functions are like the `generateText` and `streamText` helpers we explored earlier. They allow developers to integrate structured data generation into their React components while maintaining type safety. Let's analyze them briefly before we see how to integrate them in our web application.

#### GENERATEOBJECT

The `generateObject` function is used for one-time generation of structured data from an AI model. It takes a Zod schema and an AI completion function as inputs, sends a prompt to the AI model, and receives a response. The response is then validated against the provided Zod schema. If validation succeeds, it returns the structured, typed object; otherwise, it retries the generation process (up to a specified limit). To use `generateObject` with schema validation, you'll need to install Zod in your project by running `npm install zod`. The following listing shows an example.

**Listing 4.10 Generating structured responses with Zod**

```
import { z } from 'zod';
import { google } from '@ai-sdk/google';
import { generateObject } from 'ai';
```